INCH-POUND

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24 February 1994
SUPERSEDING
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MILITARY SPECIFICATION

BUTYL STEARATE, NORMAL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of N-Butyl Stearate (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Standards. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the department of Defense Index of Specifications and standards (DODISS) and supplement there to, cited in the solicitation (see 6.2).

STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 6810

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D92	Flash and Fire Points by Cleveland Open Cup
D460	Sampling and Chemical Analysis of Soaps and Soap Products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in libraries or other informational services).

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Acidity (or stearic acid).** The maximum acidity shall be 0.5 percent (see 4.3.1).

3.2 **Color.** The color shall be not darker than that of the 300 parts per million platinum cobalt standard (see 4.3.2).

3.3 **Saponification number.** The saponification number shall be from 165 to 176 (see 4.3.3).

3.4 **Iodine number.** The iodine number shall be 6.0 or lower (see 4.3.4).

3.5 **Specific gravity.** The specific gravity at 25/25°C shall be 0.855 $\pm$ 0.005 (see 4.3.5).

3.6 **Titer test.** The titer test shall be conducted at 20.5 $\pm$ 2.0°C (see 4.3.6).

3.7 **Flash point test.** The minimum flash point shall be 188°C (see 4.3.7).

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3.8 Odor. The odor shall be faint fatty, and free of butanol (see 4.3.8).

3.9 Clarity. The butyl stearate shall be free of suspended matter (see 4.3.9).

3.10 Moisture. There shall be no turbidity when 1 volume is mixed with 19 volumes of 66° gasoline (see 4.3.10).

3.11 Workmanship. The workmanship shall be in accordance with such practices as to insure compliance with all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.3.

4.2.1 Lot. A lot shall consist of material from the same batch or blending operation and subjected to the same processing operation and conditions, from one manufacturer.

4.2.1.1 Lot acceptance. The lot shall be accepted provided that all the samples satisfy the requirements of this specification when tested in accordance with the test methods of 4.3. A lot shall be rejected if one or more samples fail to satisfy these requirements.

4.2.2 Sample. A sample shall be 500 grams of n-butyl stearate, with a minimum requirement of five samples per lot. However, no more than one sample shall be selected from one shipping container unless the sample size (or number of samples) is greater than the number of shipping containers. Each sample shall be identified as to the container from which it was taken, manufacturer, plant, contract or order number, lot number, lot size, and number of containers in the lot.

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4.3 Methods of inspection.

4.3.1 Acidity. Five grams of the sample shall be accurately weighed and transferred to a 250-ml Erlenmeyer flask. In another flask, 100 ml of 95% ethyl alcohol shall be titrated to a faint pink color with 0.1N sodium hydroxide solution using phenolphthalein solution as the indicator. A 50-ml portion of the neutralized alcohol shall be transferred to the flask containing the sample, mixed thoroughly, and titrated rapidly with 0.1N sodium hydroxide solution to a pink color matching that of the remaining portion of neutralized alcohol. The acid in the sample shall be calculated to percentage stearic acid as follows:

$$Percent\ stearic\ acid = \frac{28.4\ VN}{W}$$

Where

V = ml of sodium hydroxide solution required for titration
N = normality of sodium hydroxide solution
W = weight of sample in grams

4.3.2 Determination of color.

4.3.2.1 Preparation of color standard. The stock color standard, containing 300 ppm of platinic cobalt chloride (APHA color standard), may be purchased (see 6.3) or prepared as follows: A 747-gram portion of potassium chloroplatinate (K_2PtCl_6) containing 0.300 grams platinum) and 0.600 grams of crystallized cobaltous chloride (CoCl—6HO) containing 0.149 grams of cobalt) shall be dissolved in distilled water containing 60-ml reagent grade hydrochloric acid. The solution shall be diluted to 1.000 liter with distilled water, and mixed well. This solution shall have a color intensity equivalent to 300 ppm of platinic cobalt chloride.

4.3.2.2 Determination of color. By means of a pipet, 100 ml of the 300 ppm standard shall be transferred to a 100-ml Nessler tube. A second Nessler tube shall be filled to the mark with the sample as received. The intensity of the color of the sample shall be compared with the standard by looking vertically downward through the two solutions against a white background. The sample color shall not be darker than the standard.

4.3.3 Saponification number. Two grams of the sample shall be accurately weighed and transferred by means of a weighing pipet to a 250-ml Erlenmeyer flask having a standard-taper, ground-glass neck. Exactly 25 ml of 0.5N alcoholic potassium hydroxide shall be added. The flask shall be attached to a reflux condenser by means of a ground-glass joint, and the contents boiled gently for approximately 1 hour. The condenser and the ground-glass joints shall be washed with approximately 25 ml of distilled water. The solution shall be cooled to room temperature, 10 drops of phenolphthalein indicator solution added, and the excess potassium hydroxide titrated with 0.3N hydrochloric acid. A blank shall be run concurrently with the sample. The saponification number for butyl stearate shall be calculated as follows:

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$$\textit{Saponification number} = \frac{56.1 \times (A - B)\ N}{W}$$

Where

A = ml of acid used to titrate the blank
B = ml of acid used to titrate the sample
N = normality of acid
W = weight of sample in grams

4.3.4 Iodine number. The iodine number shall be calculated in accordance with the procedures in sections 51 and 52 of ASTM D460.

4.3.5 Specific gravity. The specific gravity shall be determined at 25°C by means of a pycnometer or a hydrometer.

4.3.6 Titer test. Titer test shall be done in accordance with sections 46 and 47 of ASTM D460.

4.3.7 Flash point test. Flash point test shall be done in accordance with ASTM D92.

4.3.8 Odor. Odor shall be checked by smelling the sample.

4.3.9 Clarity. Sample shall be visually examined for clarity.

4.3.10 Moisture. Moisture shall be checked for by mixing one volume of sample with 19 volumes of 66° gasoline, then examined for any evidence of turbidity.

5. PACKAGING

5.1 Packing, Level C (domestic shipment, items for immediate use). Unless otherwise specified in the contract or order, N-butyl stearate shall be packed in standard steel or aluminum containers. The containers shall be constructed as to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall conform to Uniform Freight Classification, National Motor Freight Classification, or to rules of other carriers applicable to the mode of transportation at the lowest freight rate.

5.2 Marking. In addition to any special marking required by the contract or order, each container shall be marked in accordance with MIL-STD-129, and contain as a minimum the following information: material, specification number, quantity, manufacturer, contract number, gross weight, date, lot number.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The N-Butyl Stearate described herein is intended for use in propellant powders.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Lot size.

d. Activity that is to perform the tests of section 4.

6.3 Color standard. The stock color standard, containing 300 ppm of platinic cobalt chloride (APHA) color standard, has been found satisfactory for use in the color determination (See 4.3.2.1).

6.4 Subject term (key word) listing.

Chemicals
Saponification number
Specific gravity

Preparing activity:
Navy - OS
(Project 6810-N085)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-21465A(OS)	1994 FEBRUARY 24

3. DOCUMENT TITLE
BUTYL STEARATE, NORMAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER	b. TELEPHONE (Include Area Code) (1) Commercial: 301-743-4510/4253 (2) AUTOVON: 354-4510/4253
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE. CODE 8420 INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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